PSYC 207 Fall 2026

Exam 1 Review

Question 1 - Data & Study Design

A behavioral science research team wants to evaluate the effectiveness of different stress-management interventions in reducing perceived stress among adults. To conduct the study, 150 adults aged 18–55 years from NYC were randomly selected from individuals enrolled in a community behavioral health research program. The participants were randomly assigned to one of three intervention groups: mindfulness-based training, cognitive behavioral therapy (CBT), or stress-management education, with 50 participants assigned to each group.

The researchers measured participants' perceived stress levels before and after a four-week intervention using a standardized psychological stress scale ranging from 0 to 40, with higher scores indicating greater perceived stress. At the end of the four weeks, participants in the mindfulness-based training group had an average stress score of 18, compared with 16 among participants in the CBT group and 22 among participants in the stress-management education group. The researchers also collected information on each participant's gender (male/female), highest level of education completed (high school, bachelor's degree, or graduate degree), and average number of hours of sleep per night.

The researchers wanted to determine whether the type of intervention had an effect on participants' perceived stress levels after four weeks.

a. Identify the *sample* and the *population* of interest in the above study.

b. Classify the study method as *Descriptive, Correlational, Experimental, or quasi-experimental.* Explain your reasoning.

c. Identify any one *parameter* and its associated *statistic* in the study, stating the specific values for each if known.

d. List ***all*** the variables in the study and identify each variable as discrete or continuous.

e. Give the scale of measurement (nominal, ordinal, interval, ratio) for each variable identified in part (d) above.

Question 2 - Frequency Distributions & Central Tendency

The following scores are the ages for a random sample of $n = 32$ drivers who were issued parking tickets in Chicago during 2019. Determine the best interval width and place the scores in a grouped frequency distribution table. From looking at your table, does it appear that tickets are issued equally across age groups?

57	30	45	59	39	53	28	19
34	21	34	38	52	29	64	39
22	44	46	26	56	20	33	58
32	25	48	22	51	26	63	51

a. Create a grouped frequency table to organize the data. Use classes: 19–26, 27-34, etc and be sure to include a cumulative frequency column.

b. Describe the distribution of the data based on the frequency table. From looking at your table, does it appear that tickets are issued equally across age groups?

c. Using the **frequency table** estimate the median score. Explain how you used the table to find the median.

d. Suppose the score of 64 was a typo and should be 46. Without doing doing any calculations, explain how this change would affect the ***mean***, the ***median***, and the ***standard deviation***. In particular, state whether these statistics will increase, decrease, or stay the same and give a reason.

Question 3 - Variability

Consider a small population of 8 students with the following scores on a recent exam:

57 30 45 95 39 53 28 20

a. Calculate the standard deviation for the scores and interpret its meaning in context of the scores. Be sure to show all mathematical work.

b. Calculate the first (Q_1), second (Q_2), and the third quartiles (Q_3) for the scores.

c. Use the $1.5 \times IQR$ rule to determine if there are any suspected outliers among the eight scores.

d. Sketch a box plot to visualize the distribution of the scores. Be sure to use an appropriate scale so your plot is reasonable.

Question 4 - Z-Scores and Central Limit Theorem

A researcher is interested in the effects of a "smart drug" on performance on a standardized intelligence test. It is known that the population mean and standard deviation for this test are $\mu = 200$ and $\sigma = 45$ respectively.

a. Emma, a participant who receives the drug subsequently earns a score of $X = 220$. Should the researcher be convinced that Emma's score is appreciably different from the mean? Explain your answer.

b. Eric, who also took the drug, scored $X = 198$ on another version of the test. The population mean and standard deviation on this version are $\mu = 180$ and $\sigma = 30$ respectively. Who between Emma and Eric performed better relative to their respective populations? Explain your answer.

c. Recall that a **sampling distribution for the mean** is the set of all possible samples of size n and their means. For a sampling distribution of size $n = 49$ test takers (original test version with $\mu = 200$ and $\sigma = 45$), what would be the *expected mean* and *standard error*? Sketch and label the distribution.

d. Suppose the researcher takes a random sample of $n = 49$ people and gives them the drug followed by the test. The sample mean test score for these 49 people is found to be $M = 215$. Would you consider this mean to be unusual? Justify your answer.